

Asadullah Farooqi (Asadullah)

+923111686184, asadullah.itcgcs@gmail.com, [GitHub](#), [LinkedIn](#), [Portfolio](#), Located: **Islamabad, Pakistan**

Languages: **English** (Fluent), **Urdu** (Fluent), **Arabic** (Good in Grammar), **Persian** (Beginner), **Pashto** (Fluent)

Summary: Software engineer with 5+ years of experience developing scalable and robust applications. I've worked in embedded systems, IoT, Web Development, automation, and Fintech. Enthusiastic about Fleet-reliability, Linux, performance, automation, distributed systems, Rust, and Elixir

SKILLS

Languages: Golang, Python, Rust, Bash | **Infrastructure:** Kubernetes, Docker, AWS (EKS, RDS, SQS, SNS, Aurora etc), Azure IoT | **Backend:** MySQL, PostgreSQL, MongoDB, Redis, Elasticsearch, ETL Pipelines | **Protocols:** gRPC, REST, GraphQL, MQTT, HTTP | **Tools:** Go-kit, Gin, Echo, Django, Flask, Tokio, Git, GitHub Actions

Permanent and Full-Time

Sr Software Engineer at [SafePay](#) Apr 2022 - Present

- Assist new engineers with architecture decisions and mentoring junior engineers on production systems
- **Eliminated production bugs to 0% over 12 months** by implementing distributed locking mechanism preventing concurrent payment capture requests
- **Automated 99% of manual reconciliations** by implementing controllable async golang service.
- Improved analytics by 100% and controlled the RDS cost growth through Elasticsearch integration across microservices ecosystem with event listeners and backfillers
- To achieve faster backfill for 5+ million records without choking the DB or increasing cost I wrote efficient jobs using worker pool concurrency model to sync PostgreSQL and Elasticsearch from primary database
- Designed and implemented complete data warehouse infrastructure (PostgreSQL) including ETL pipelines, Kubernetes batch jobs, and SQS event streaming
- Developed advanced settlement generation system supporting marketplace models (Stripe Connect architecture) for complex multi-party payouts
- Wrote utility billing service covering 90% of national service providers with high availability requirements
- Created new payment gateway integrations (Easypaisa, AbhiPay, card payments) as Golang microservices using Go-kit, gRPC, with optimized MySQL schemas
- Implemented audit logging system for admin operations ensuring compliance and full traceability
- Extended core services with security layers: BIN-based MDR charging, validation middleware, gRPC clients, and Protocol Buffer definitions
- Developed payment reporting APIs and complex SQL queries (MySQL/PostgreSQL) powering real-time merchant and admin dashboards
- Tech: Golang, Kubernetes, Docker, AWS (SQS, SNS, RDS, EKS), MySQL, PostgreSQL, Elasticsearch, gRPC, HTTP, ETL

Software Engineer at [Siemens DISW](#) Feb 2020 - Mar 2022

- Integrated AWS IoT Core and Azure IoT Hub into edge solution for MQTT telemetry, device provisioning, authentication, and OTA firmware updates. Also added support for Flogo (no-code solution)
- Developed features for embedded IoT framework in C for resource-constrained devices
- Built comprehensive Python test automation framework from scratch for unit, integration, and firmware update testing, reducing manual QA cycles
- Tech: C, Golang, Python, AWS IoT, Azure IoT, Docker, Linux, Make, CMake, MQTT, GCC/Delve Debuggers

Internships, Part-Time, and Short-Term Contracts

Software Engineer Intern at [Maher Inc.](#) Dec 2018 - Mar 2019

- Fixed UI problems and bugs in [StyleMepretty](#).
- Added APIs for statistics in the backend and charts in the front end to display various matrices. My improvements made the dashboard more analytical, providing clean data and different types of charts.
- Tech-stack: Python, Django, AngularJS, ReactJS, HTML, CSS, Echart JS, Pandas, and Docker.

First Project [Inventory web app](#) Nov 2017 - Mar 2018

- Built an inventory-responsive web app for a local businessman to manage branches and ledgers. As my first project ever, I learned a lot. Tech: Python, Django, [Jinja](#), Bootstrap, [Pythonanywhere](#), Unicorn, WSGI.

Software development Contractor with **Various Clients** Dec 2019 - Feb 2023

- Added features for [BP8 Fitness](#) using Golang, React, MongoDB, Docker, and onboarding joiners.
- Resolved the race condition issues in the API.
- The [Halidai](#) app was a social media app. I wrote a Django web app that used a CDN for media file storage and MongoDB for data management.
- For [Punch Group](#), I implemented the checkout frontend for [Skipify](#) using NextJS.
- Wrote a Flask web app for the sales team to interact with the lead generation, which automated the manual workflow of file generation handovers. Also optimized the lead generation Scrapers to reduce the cloud cost.

EDUCATION

Government College Loralai	FSc (Mathematics)	2013 - 2015
Farooqi Public High School	Matriculation (Science)	2011 - 2013